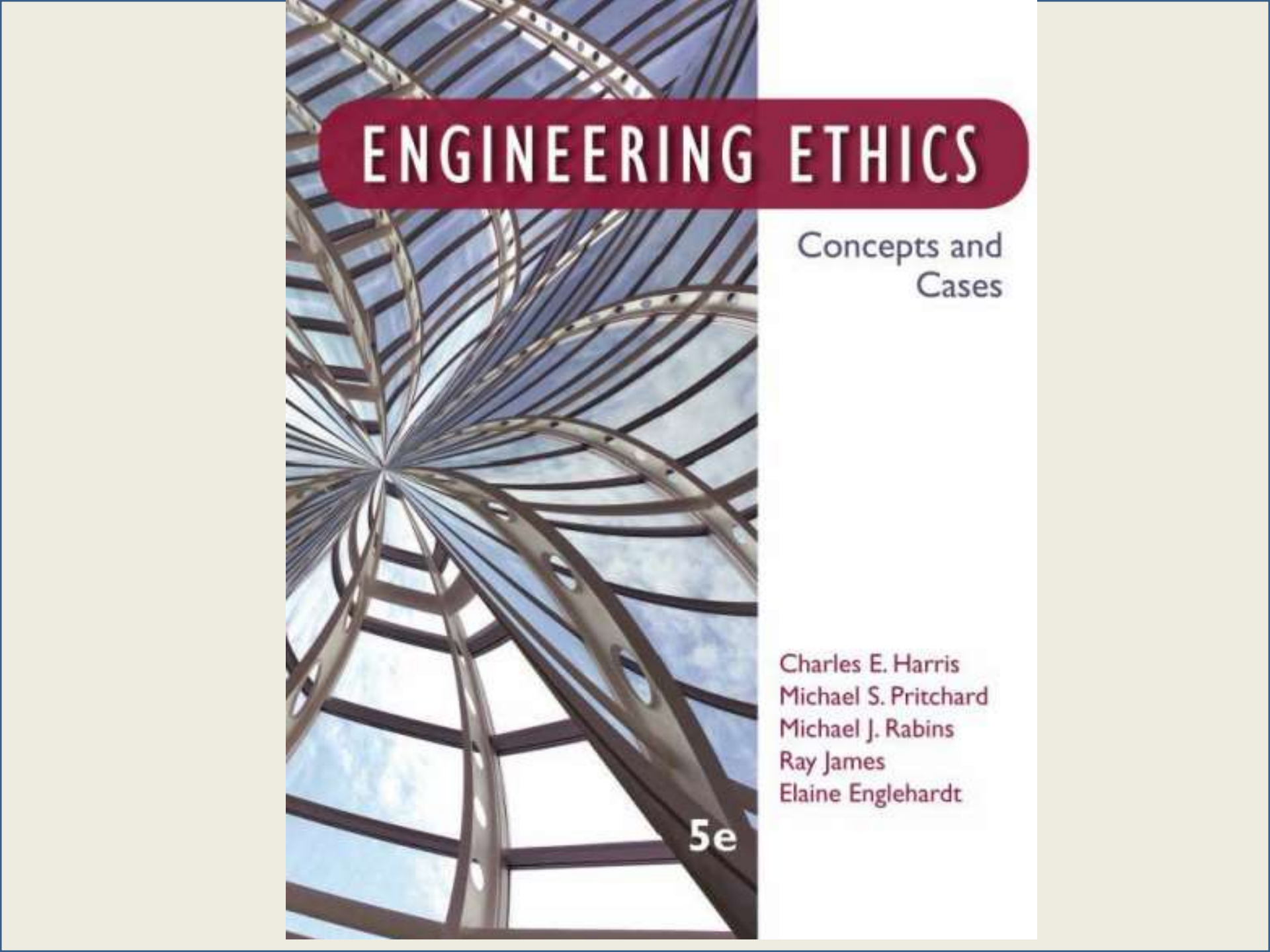


Introduction of Engineering, professionalism, ethics



ENGINEERING ETHICS

Concepts and
Cases

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What is Engineering, professionalism?

Engineering is the application of scientific and mathematical principles to design, build, and maintain structures, machines, systems, and processes that meet specific needs and criteria. Professionalism in engineering refers to the conduct, ethics, and values that guide engineers in their work and interactions with society, emphasizing competence, integrity, and a commitment to public safety and welfare.

Engineering:

- Involves using scientific and technical knowledge to solve practical problems.**
- Includes the design, construction, and maintenance of various systems and technologies.**
- Requires a deep understanding of natural laws, physical resources, and engineering principles.**
- Aims to create solutions that are safe, efficient, and beneficial to society.**



Professionalism:

- Encompasses ethical conduct, technical competence, and a commitment to public safety.**
- Involves adhering to a code of ethics, maintaining integrity, and acting responsibly.**
- Requires engineers to stay updated with the latest knowledge and advancements in their field.**



Professionalism:

- **Emphasizes the importance of teamwork, communication, and collaboration.**
- **Includes a focus on public approbation and acting in the best interests of society.**
- **Involves exercising judgment and discretion in engineering practice.**
- **Requires engineers to be accountable for their work and decisions.**



Responsibilities of Computer science and engineering students:

Computer Science and Engineering (CSE) students are responsible for learning the fundamental principles of computing and engineering, and applying them to design, develop, and implement computer systems and software. This includes understanding hardware and software components, algorithms, data structures, and various programming languages. They also develop problem-solving, critical thinking, and teamwork skills to contribute to the field of computer science and engineering.

Responsibilities of Computer science and engineering students:



Core Responsibilities

Understanding Fundamental Concepts:

CSE students need to grasp the core principles of computer science and engineering, including algorithms, data structures, programming languages, computer architecture, and operating systems.

Developing Software:

This involves designing, coding, testing, and debugging software applications, ranging from simple programs to complex systems.

Core Responsibilities

Designing Hardware:

Students explore the design and development of computer hardware components, such as microprocessors, circuit boards, and memory devices.

Problem-Solving:

They learn to analyze complex computational problems and develop effective solutions using their knowledge of algorithms, data structures, and programming.

Core Responsibilities

Research and Innovation:

CSE students are encouraged to explore emerging technologies, conduct research, and contribute to advancements in the field.

Collaboration:

They work in teams, participate in projects, and learn to communicate effectively with other engineers and stakeholders.

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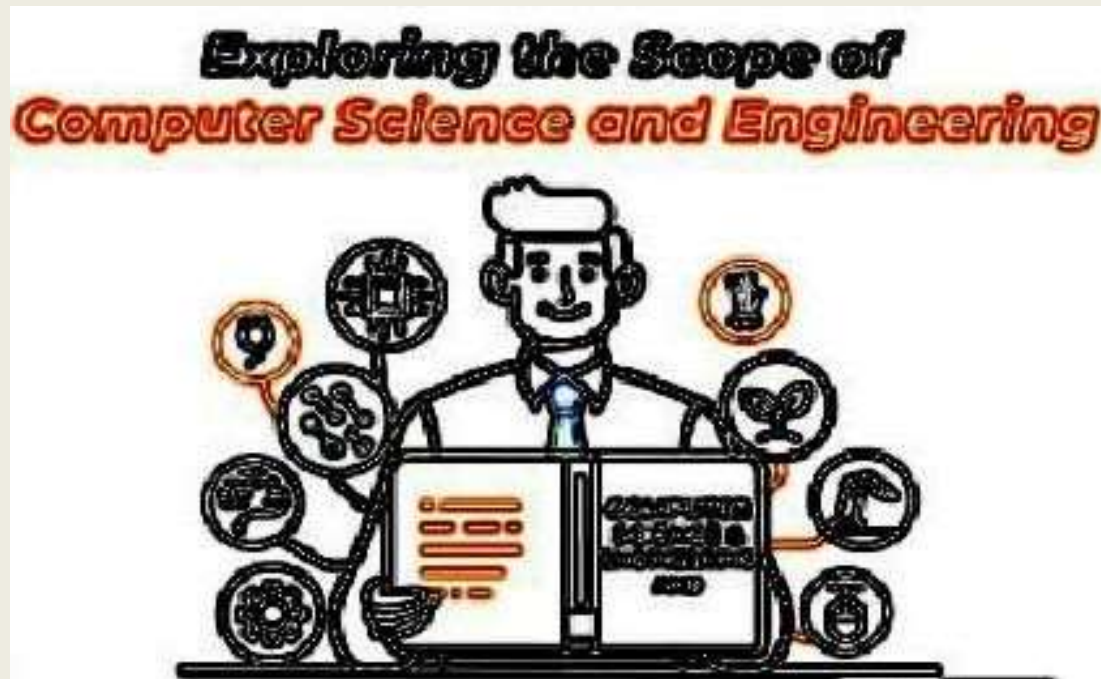
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Core Responsibilities

Staying Updated:

Keeping up with the latest advancements in technology and continuously expanding their knowledge and skills is crucial.



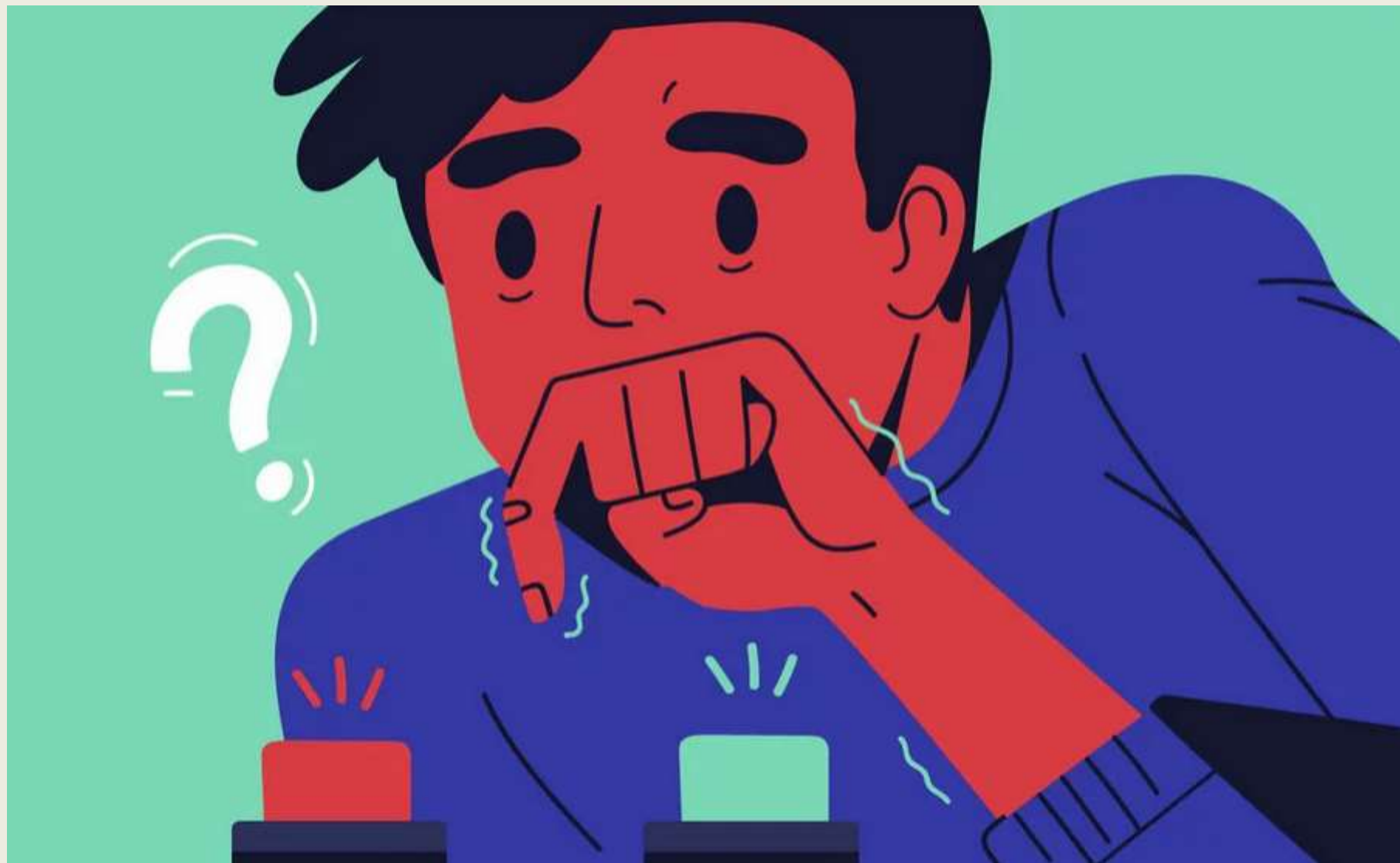
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Morals of computer science and engineering

The morals of computer science and engineering center around responsible innovation and the ethical use of technology. Key principles include respecting user privacy, ensuring system integrity, promoting fairness and equity, and considering the broader societal impact of computing systems. These principles guide professionals to develop and deploy technology that benefits society while minimizing harm.



Core Principles:

Respect for Privacy:

Protecting user data and information from unauthorized access, use, or disclosure is paramount. This includes safeguarding personal information and respecting the confidentiality of communications.

System Integrity:

Ensuring the reliability, security, and stability of computer systems and software is crucial. This involves preventing unauthorized access, malicious attacks, and data corruption.

Core Principles:

Fairness and Equity:

Striving to develop and deploy technology that is accessible to all and avoids perpetuating biases or discrimination. This includes addressing issues of access, affordability, and the potential for technology to exacerbate existing inequalities.

Social Responsibility:

Considering the broader societal impact of computing systems, including their potential effects on employment, the environment, and social structures. This involves anticipating potential harms and working to mitigate them.

Core Principles:

Intellectual Property:

Respecting copyright, patents, and other forms of intellectual property. This includes avoiding plagiarism, respecting licenses, and properly attributing the work of others.



**BUSINESS
INTELLIGENCE**



**COMPUTER
VISION**



**ARTIFICIAL
INTELLIGENCE**



**BIG
DATA**



**NATURAL
LANGUAGE PROCESSING**



**MACHINE
LEARNING**



**DATA
MINING**



**DATA
VISUALIZATION**



**DATA
ETHICS**

Specific Considerations:

Software Piracy:

Refraining from copying or distributing copyrighted software without permission.

Free Expression and Content Control:

Navigating the complexities of regulating online content, balancing free speech with the need to prevent harm and misinformation.

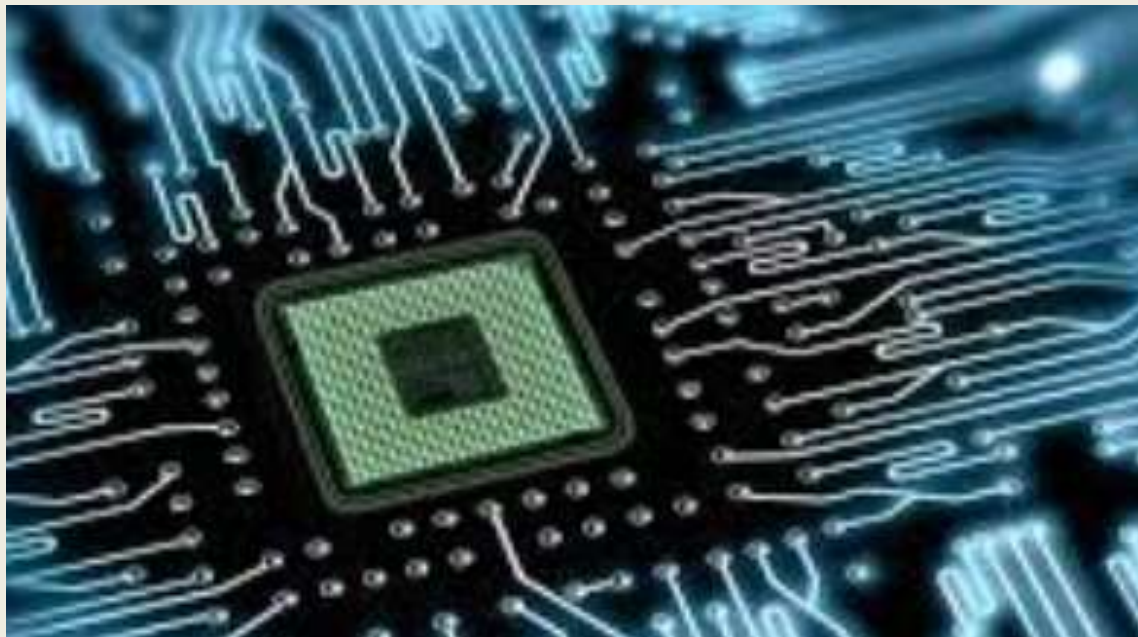
Artificial Intelligence Ethics:

Addressing the ethical implications of AI, including bias in algorithms, autonomous decision-making, and the potential for job displacement.

Specific Considerations:

Data Security and Privacy:

Protecting sensitive information in an increasingly data-driven world, including personal data, financial information, and medical records.



Professional Codes of Conduct:

Many professional organizations, such as the [ACM \(Association for Computing Machinery\)](#) and the [IEEE Computer Society](#), have established codes of ethics to guide the conduct of their members. These codes often emphasize principles like integrity, honesty, public safety, and the importance of professional judgment. By adhering to these moral principles and codes of conduct, computer scientists and engineers can contribute to a future where technology is used to empower individuals, improve society, and promote the common good.

Ethics of computer science and engineering :

Computer ethics in computer science and engineering refers to the moral principles that guide the responsible use and development of computer technology. It encompasses issues like privacy, security, intellectual property, and the societal impact of computing systems. Essentially, it's about ensuring technology is used ethically and responsibly to benefit individuals and society as a whole.

Key Areas of Computer Ethics::

Privacy:

Protecting personal information and ensuring data is handled responsibly and securely.

Security:

Preventing unauthorized access, misuse, or damage to computer systems and data.

Intellectual Property:

Respecting copyrights, patents, and other forms of intellectual property related to software and hardware.

Key Areas of Computer Ethics::

Social Impact:

Considering the broader consequences of technology on society, including issues like bias, accessibility, and the digital divide.



Ethical Principles for Computing Professionals:

Do no harm: Avoid creating or using technology that could harm individuals or society.

Be honest and trustworthy: Ensure accuracy, avoid deception, and maintain integrity in all computing activities.

Respect the rights of others: Uphold privacy, intellectual property, and other fundamental rights.

Promote fairness and justice: Strive to create technology that benefits all members of society and avoids discrimination.

Be accountable: Take responsibility for the consequences of your work and the technology you develop.

Examples of Ethical Dilemmas in Computing:

Algorithmic Bias:

When an algorithm reflects and amplifies existing societal biases, leading to unfair or discriminatory outcomes.

Data Privacy:

Concerns about the collection, storage, and use of personal data, especially in the age of big data and artificial intelligence.

Examples of Ethical Dilemmas in Computing:

Autonomous Systems:

Ethical considerations surrounding the development and deployment of self-driving cars, robots, and other AI-powered systems.

Surveillance Technologies:

The use of technology for surveillance and monitoring, raising concerns about privacy and freedom.

Organizations and Resources:

ACM (Association for Computing Machinery):

The world's largest educational and scientific computing society, with a code of ethics for computing professionals.

IEEE (Institute of Electrical and Electronics Engineers):

An organization that develops standards and provides resources for engineers, including a code of ethics.

Examples in Scenarios:

Computer ethics scenarios often involve dilemmas that require individuals to weigh different ethical considerations and make difficult decisions. These scenarios can be used in educational settings to help students develop their ethical reasoning skills. For example:

Examples in Scenarios:

- A programmer discovers a security vulnerability in a widely used software application but is hesitant to report it due to concerns about the potential impact on the company and its users.
- A data scientist working for a social media company faces pressure to use personal data in ways that could be considered invasive or manipulative.
- An engineer working on autonomous vehicles grapples with the ethical implications of designing a system that could make life-or-death decisions in an accident.